POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor

Semester: Spring

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Explain the Microprocessor based System with basic block diagram. 5

b) Why the speed of Von-Neumann architecture is low than Harvard architecture? Explain Von-Neumann architecture with various registers used in it. 5

c) Why the flag is also called PSW? Explain with the help of 8085 flag register. 5

a) What are the different rotate instructions used in 8085 instruction set? Explain with the help of a simple example for each of them. 7

b) What is the difference between a counter and a time delay loop? Explain with the help of a flowchart. 8

a) Classify the Bus on the basis of their function. Differentiate between Synchronous and Asynchronous Bus. 7

b) What do you mean by address decoding? Design a scalable address decoding circuit to interface 64-byte RAM memory and 128-byte ROM memory. 8

a) With a suitable diagram explain the write operation with a DRAM cell. 7

b) Write an assembly language program in 8086 which reads a string from the keyboard and then displays the string in the reverse order. For example if the input string read from the keyboard is "POKHARA UNIVERSITY", the output should be "YTISREVINU ARAHKOP". 8

a) Define addressing mode. Explain the addressing modes of 8086 with proper examples. 7

b) What is the significance of interrupt in microprocessor? Explain the IVT with suitable figure. 8

6. a) Explain EU and BIU with the help of well labelled block diagram of 8086 microprocessors. 7

   b) Differentiate between synchronous and asynchronous serial data communication. Also explain briefly the protocols of synchronous serial data communication. 8

7. Write short notes on: (Any two) 2×5

   a) RTL Instruction Descriptions

   b) Memory mapped I/O Vs I/O mapped I/O

   c) Serial data transmission standards

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